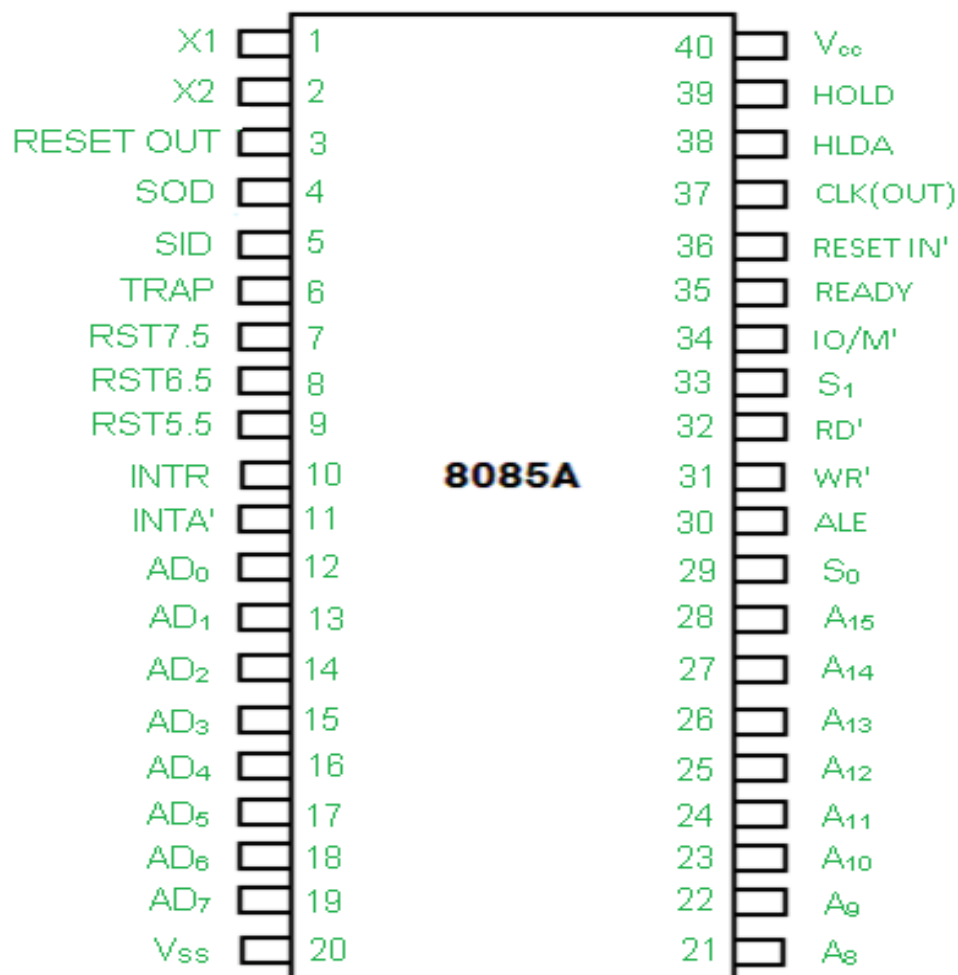


Microprocessor - 8085 Pin Configuration

The following image depicts the pin diagram of 8085 Microprocessor –



The pins of a 8085 microprocessor can be classified into seven groups –

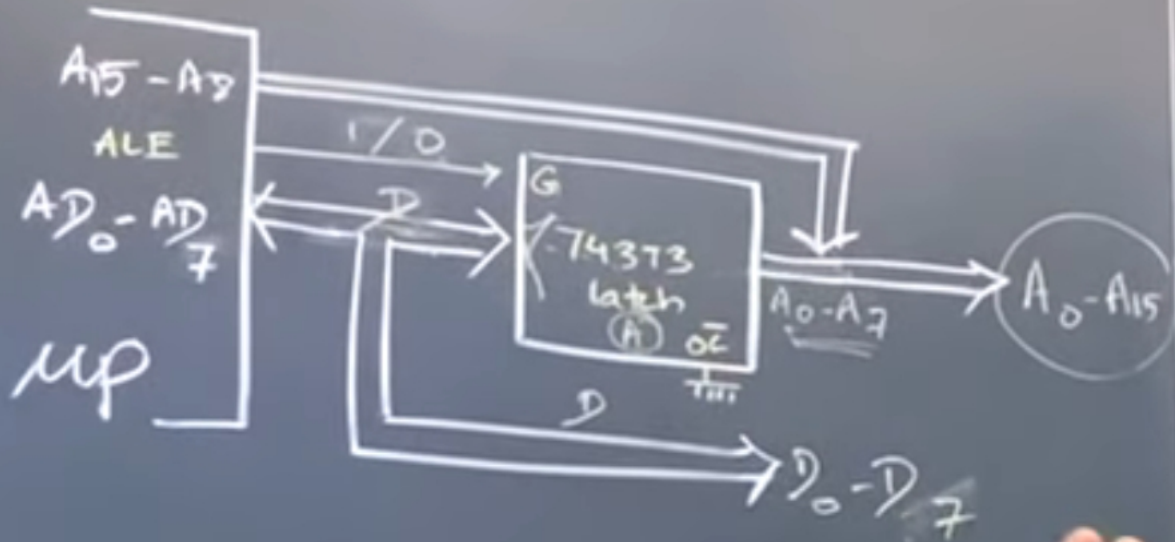
Address bus

A15-A8, it carries the most significant 8-bits of memory/IO address.

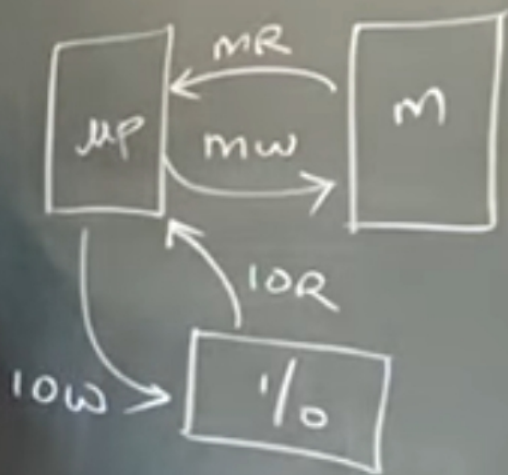
Data bus

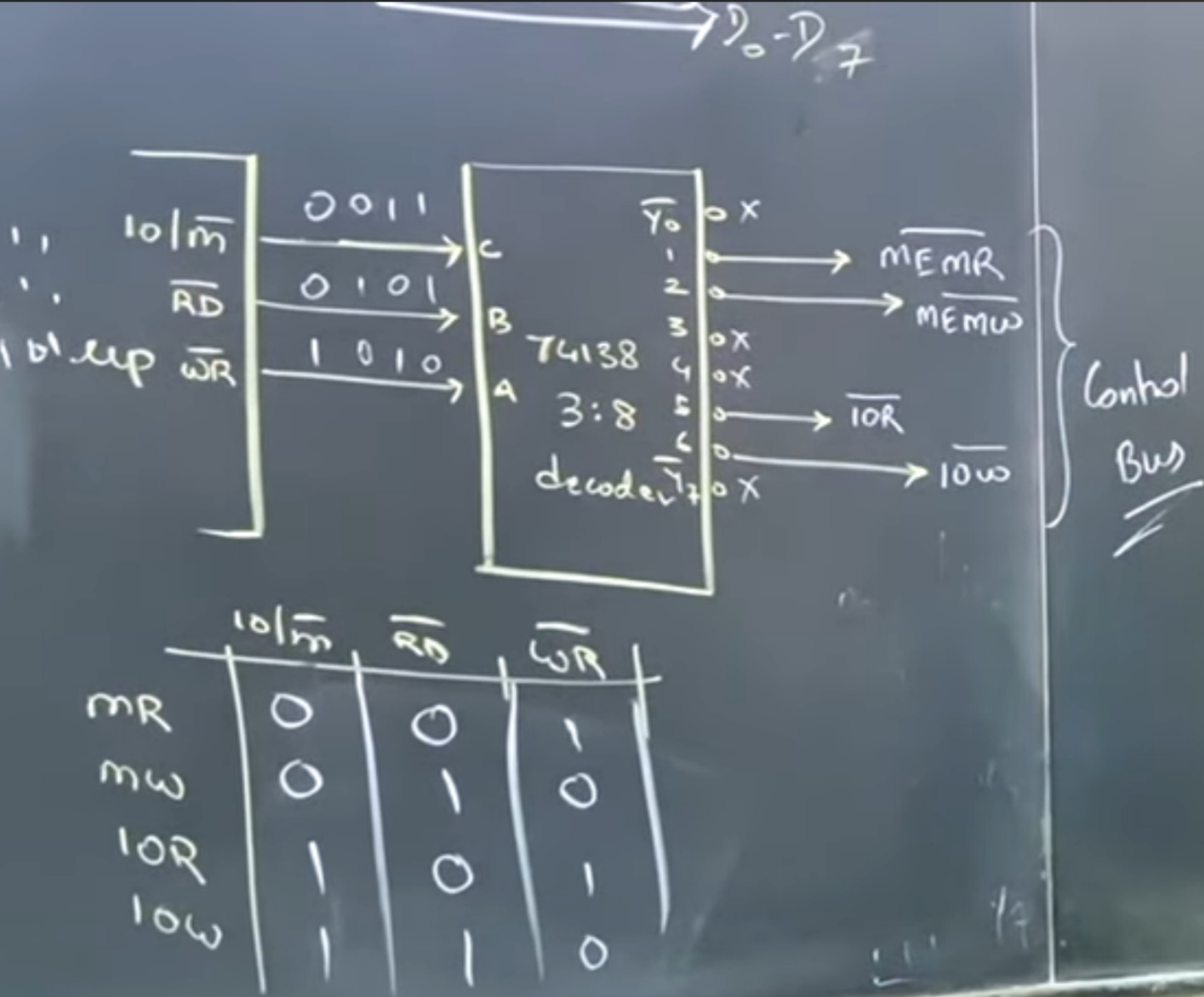
AD7-AD0, it carries the least significant 8-bit address and data bus.

Control and status signals

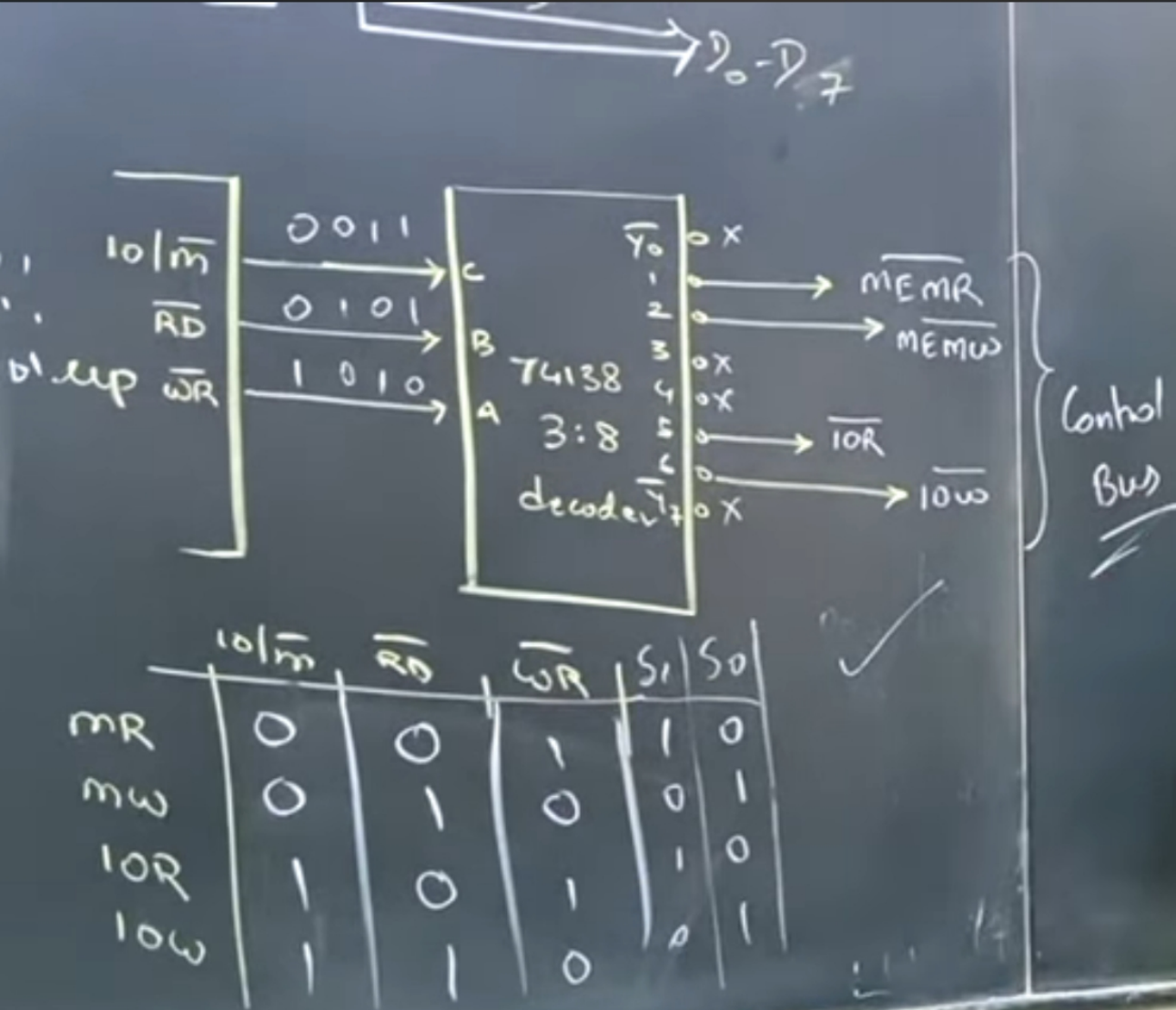


tion





The **outputs** of the 74138 decoder are **ACTIVE LOW** signals which means that when given output is selected its value would be 0 (GND).



This isn't a fact but to understand the signals :

$S_1 \Rightarrow$ S-Input

$S_0 \Rightarrow$ S-Output

When μp gets input(read) from $IO/\overline{M}$, $S_1 = 1$

When μp sends output(write) to the $IO/\overline{M}$, $S_0 = 1$

These signals are used to identify the nature of operation. There are 3 control signal and 3 status signals.

Three control signals are RD, WR & ALE.

- **$\overline{RD}$** – This signal indicates that the selected IO or memory device is to be read and is ready for accepting data available on the data bus.
- **$\overline{WR}$** – This signal indicates that the data on the data bus is to be written into a selected memory or IO location.
- **ALE** – It is a positive going pulse generated when a new operation is started by the microprocessor. When the pulse goes high, it indicates address. When the pulse goes down it indicates data.

Three status signals are IO/M, S0 & S1.

IO/ $\overline{M}$

This signal is used to differentiate between IO and Memory operations, i.e. when it is high indicates IO operation and when it is low then it indicates memory operation.

S1 & S0 (Status Signals)

These signals are used to identify the type of current operation.

Power supply

There are 2 power supply signals – VCC & VSS. VCC indicates +5v power supply and VSS indicates ground signal.

Clock signals

There are 3 clock signals, i.e. X1, X2, CLK OUT.

- **X1, X2** – A crystal (RC, LC N/W) is connected at these two pins and is used to set frequency of the internal clock generator. This frequency is internally divided by 2.
- **CLK OUT** – This signal is used as the system clock for devices connected with the microprocessor. **μp shares the same clock, it receives from crystal oscillators**

Interrupts & externally initiated signals

Interrupts are the signals generated by external devices to request the microprocessor to perform a task. There are 5 interrupt signals, i.e. TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR. We will discuss interrupts in detail in interrupts section.

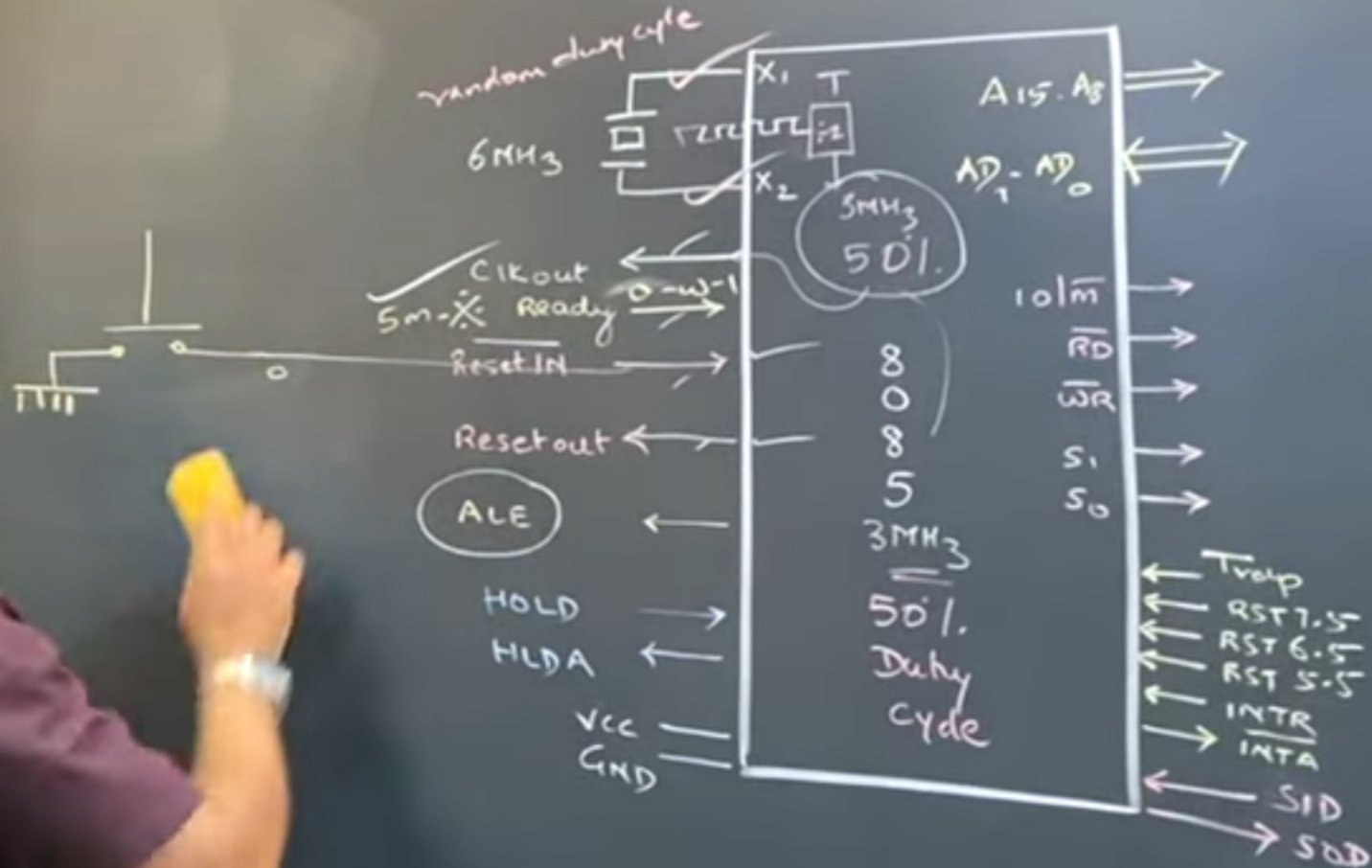
- **INTA** – It is an interrupt acknowledgment signal. **Only triggered in case of INTR**
 - **RESET IN** – This signal is used to reset the microprocessor by setting the program counter to zero.
 - **RESET OUT** – This signal is used to reset all the connected devices when the microprocessor is reset.
 - **READY** – This signal indicates that the device is ready to send or receive data. If READY is low, then the CPU has to wait for READY to go high.
 - **HOLD** – This signal indicates that another master is requesting the use of the address and data buses.
 - **HLDA (HOLD Acknowledge)** – It indicates that the CPU has received the HOLD request and it will relinquish the bus in the next clock cycle. HLDA is set to low after the HOLD signal is removed.
- HOLD & HLDA are DMA(Direct Memory Access) signals**

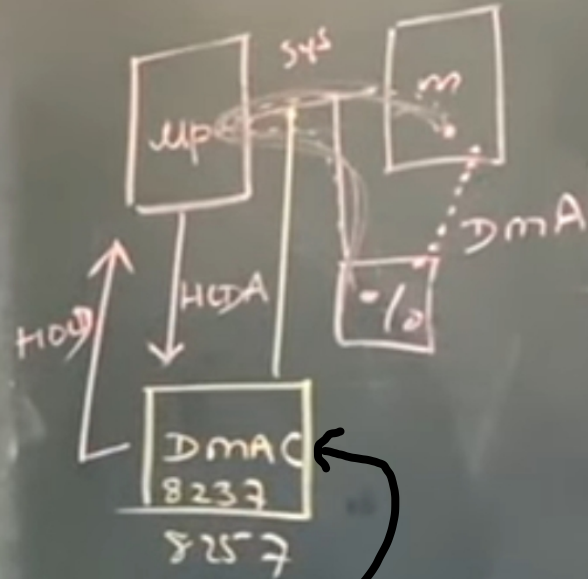
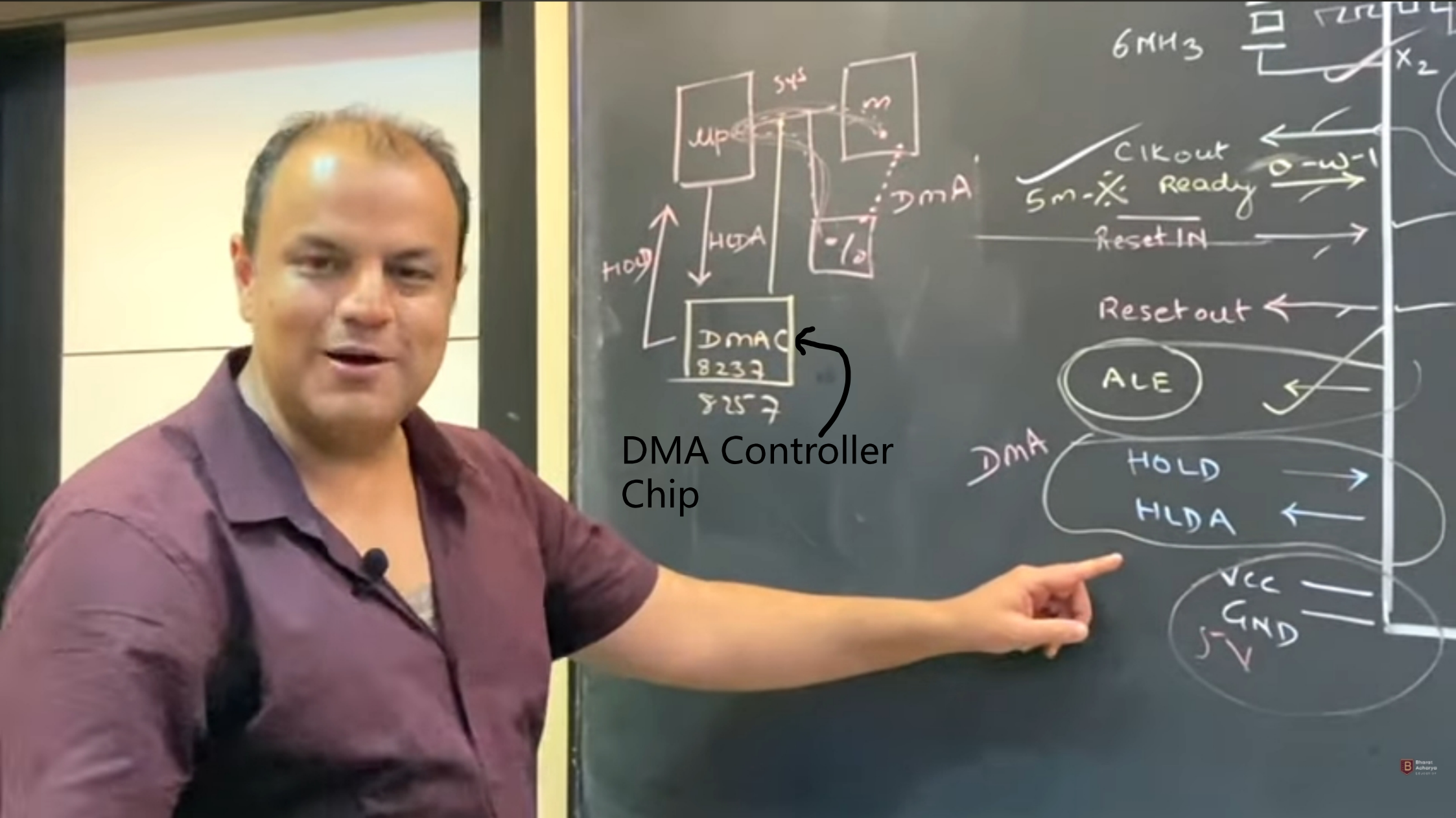
Serial I/O signals

There are 2 serial signals, i.e. SID and SOD and these signals are used for serial communication.

- **SOD** (Serial output data line) – The output SOD is set/reset as specified by the SIM instruction.
- **SID** (Serial input data line) – The data on this line is loaded into accumulator whenever a RIM instruction is executed.

READY = Synchronizes the μp with slower peripheral devices. When it's 1 - μp sends data, when it's 0 - μp enters Wait State and constantly keep checking until it gets 1 to send data again.





DMA Controller Chip

